Angular momentum deficit and exponentially long collision safety in the planetary problem

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Abstract

Laskar and Petit conjectured that a planetary system whose angular momentum deficit (AMD) is too small to allow orbits to touch is, away from short-period resonances, stable in practice. We prove a rigorous asymptotic form of the mechanism behind this conjecture, in the plane and in space, up to the sharp strong threshold, which also forbids a planet from reaching the star. Fix the mass ratios, a compact set of ordered semi-major axes and a margin $\delta > 0$ below the threshold. If the planetary masses are of order μ, with μ small, and the initial mean motions are Diophantine with exponent τ, then for $|t| \le T_0 \exp(c\mu^{-1/(2(\tau+1))})$ the semi-major axes move by $O(\sqrt{\mu})$, the AMD stays at least $\delta/2$ below the threshold, and the orbital annuli stay ordered and uniformly separated. Without arithmetic conditions the same holds with Nekhoroshev's exponents. Exact conservation of angular momentum confines the AMD while the semi-major axes drift exponentially slowly; the secular dynamics may be chaotic. The constants are not explicit. The results under the Diophantine condition are formally verified in the Lean proof assistant.

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1 Introduction

Consider a star and $n \ge 2$ planets whose masses are a small multiple μ of the mass of the star, interacting by Newtonian gravity. To first approximation each planet follows a Kepler ellipse. The

mutual attractions deform these ellipses: over times of order μ^{-1} the eccentricities, inclinations and orientations change appreciably (the secular evolution), while the semi-major axes stay nearly constant. Whether the orbits can come to intersect, so that close encounters, collisions or ejections become possible, depends on how far the eccentricities can grow. In this paper we prove that an elementary conserved quantity bounds this growth for an exponentially long time, with a threshold that is sharp.

The angular momentum deficit. The *angular momentum deficit* (AMD) of a planetary system is the difference between the angular momentum that the planets would have on circular, coplanar orbits with the same semi-major axes and their actual angular momentum [6, §2.2]. With Delaunay actions $\Lambda_k = m_k\sqrt{Gm_0a_k}$ (where m_0 is the mass of the star), eccentricities e_k and inclinations i_k relative to the plane orthogonal to the total angular momentum, it reads $\sum_k \Lambda_k(1-\sqrt{1-e_k^2}\cos i_k)$ [6, (11)]. The AMD vanishes exactly for circular coplanar prograde orbits, and a planet can reach large eccentricity only if the AMD is large enough. Laskar and Petit [6, §3] computed the least AMD at which the orbits of two planets with given semi-major axes and masses can touch. They call a system *AMD-stable* if its AMD is below this value for every adjacent pair, and *strongly* AMD-stable if, moreover, it is below the value Λ_1 at which the innermost planet can reach the star [6, §5.3]. Since the AMD is conserved by the secular dynamics at all orders, they formulate the following conjecture [6, §3]:

> *we conjecture that in absence of short period resonances, the AMD-stability ensures the practical long-term stability of the system.*

Here "practical" refers to times that are very long compared with the expected lifetime of the star [6, §3, footnote 1]. The conjecture is informal: it concerns real systems with actual masses and astronomical time scales, and it is used in [6] to classify observed multi-planet systems.

Chaos and collision safety. The secular dynamics of a planetary system need not be regular, and the AMD-stability criterion does not ask for its stability. It asks only that the eccentricities and inclinations, however they evolve, never reach a configuration in which orbits touch. In the full Newtonian problem this rests on two facts. First, the total angular momentum L is conserved exactly, not only in an averaged system. In the normalization of Section 2, where the AMD is $C = \sum_i \Lambda_i - |L|$, this gives

$$C(t) - C(0) = \sum_{i=1}^{n} \big(\Lambda_i(t) - \Lambda_i(0)\big)\,,$$

so the AMD can change only as much as the Delaunay actions $\Lambda_i = \nu_i\sqrt{a_i}$, which are functions of the semi-major axes alone. Second, the actions Λ_i are conjugate to the fast angles (the mean longitudes), and averaging over these angles shows that they drift very slowly, for a time that is exponentially long in a negative power of μ. This second fact needs no information about the slow (secular) variables, beyond the requirement that they stay in a region where the Hamiltonian is analytic and the orbits are separated. The AMD, which bounds the eccentricities through $C \geq \sum_i \Lambda_i(1-\sqrt{1-e_i^2})$, keeps the slow variables in such a region. The two facts therefore close a bootstrap: as long as the axes have moved little, the AMD has moved little, so the orbits are still separated and averaging still applies. The secular motion inside the allowed region can be chaotic.

Results. We make this mechanism rigorous, in the asymptotic regime $\mu \to 0$, up to the sharp strong threshold. Let $B(\Lambda)$ be the least of the pair thresholds $B_{i,i+1}(\Lambda)$ and of the actions Λ_i, computed exactly from the actions $\Lambda = (\Lambda_1, \dots, \Lambda_n)$; the inequality $C < B(\Lambda)$ is the strong AMD-stability condition of Laskar and Petit (Section 2). Fix the mass ratios, a compact set $\mathcal{A}$ of ordered semi-major axes and a margin $\delta > 0$. Informally, our results are as follows.

- *Theorem A (sharp barrier).* Every state with axes in a compact set of the chamber and AMD at most $B(\Lambda) - m$ has its perihelia bounded below and adjacent orbital annuli separated by a fixed distance, uniformly. No secular or dynamical information enters.
- *Theorems B and C (exponential collision safety).* In the plane and in space, if the initial axes lie in $\mathcal{A}$, the initial AMD is at most $B(\Lambda(0)) - \delta$, and the initial mean motions satisfy a Diophantine condition with exponent τ, then the motion exists and is unique for $|t| \le T_0 \exp(c\mu^{-1/(2(\tau+1))})$. During this time the AMD stays below $B(\Lambda(t)) - \delta/2$, the axes move by $O(\sqrt{\mu})$, and the orbital annuli of the planets remain strictly ordered and uniformly separated from each other and from the star, both for the canonical elements and for the osculating ellipses computed from the actual velocities. No hypothesis on the inclinations is made in space.
- *Corollary D.* For $\tau = n - 1/2$, which includes almost every initial frequency vector, the time is $T_0 \exp(c\mu^{-1/(2n+1)})$.
- *Theorem E (no arithmetic hypothesis).* Without any condition on the frequencies, the same conclusions hold for $|t| \le T_0 \exp(c\mu^{-\mathfrak{a}_n})$, with drift $O(\mu^{\mathfrak{b}_n})$, where $\mathfrak{a}_n = 2/(6n^2 - 3n + 14)$ and $\mathfrak{b}_n = 3/2\,\mathfrak{a}_n$ are Nekhoroshev's exponents for the planetary problem [7, (12.4)–(12.5)].

In all results the constants depend only on the mass ratios, $\mathcal{A}$, δ (and γ, τ), and are chosen before μ and the initial state. The results are uniform over all eccentricities, inclinations and phases allowed by the AMD condition.

Scope. These theorems are a rigorous asymptotic form of the mechanism behind the Laskar–Petit conjecture: for every fixed margin below the strong threshold, collision safety holds for exponentially long times once the masses are small enough. They do not resolve the conjecture itself, which concerns practical time scales and actual masses. Our constants are obtained by compactness and are not explicit, and we make no statement about the Solar System or any observed system. We also prove nothing about the stability of the secular dynamics.

Ideas of the proofs. The barrier (Theorem A) follows from the identity for C, the inequality $C \ge \sum_i \Lambda_i(1 - \sqrt{1 - e_i^2})$ and the continuity of B on the chamber. For Theorems B and C we work in regular Poincaré variables, in which the eccentricities and inclinations are Cartesian slow coordinates, and C is half the squared norm of the slow variables. We average over the fast angles about the linear Hamiltonian with the initial frequencies, treating the nonlinear Kepler term on an action ball of radius $\sqrt{\mu}$ as part of the perturbation. The point of the construction is that the entire set of slow variables allowed by the AMD condition, a ball, is a single domain of analyticity with a fixed complex width. The normal form is of the type of Pöschel's lemma [10, p. 192]; the Diophantine condition with radius $\sqrt{\mu}$ yields the exponent $1/(2(\tau+1))$, and a first-exit argument closes the bootstrap described above. Theorem E instead applies Nekhoroshev's theorem with slow variables [7, Theorem 4.4] on finitely many cells covering the sharp sublevel set.

Relation to earlier work. The mechanism is not new. Nekhoroshev's planetary theorem [7, §12, Theorem 12.3] concerns the *domains of planetary motions*, defined by boxes $\alpha_i \le a_i \le \beta_i$ of semi-major axes and a lower bound $|N| \ge \gamma$ on the total angular momentum of the unperturbed system, where γ exceeds the largest value for which the boundary of the domain contains collisions [7, §12.2]. For initial data in such a domain he proves that no collision occurs for a time of order $\mu^{-1}\exp(c\mu^{-\mathfrak{a}_n})$, during which the axes change by $O(\mu^{\mathfrak{b}_n})$ [7, (12.3)–(12.5)]. Theorem E has the same exponents; its hypothesis is the pointwise strong AMD condition $C(0) \le B(\Lambda(0)) - \delta$ at the initial axes, rather than a threshold uniform over a box, and its conclusions add the ordering of the orbital annuli and the osculating ellipses. Niederman [8] extended Nekhoroshev theory to the planetary problem; as described in [4, §1.3], he proved that the actions conjugate to the fast angles are stable over exponentially long times provided the secular actions remain in a bounded region, and that near circular coplanar motions the conservation of angular momentum guarantees this; Barbieri and Niederman report that his explicit estimates for the planetary three-body problem near a periodic torus give stability for a time comparable with the age of the Solar System when the mass ratio is below 10^{-13} [3, p. 2]. Barbieri and Niederman [3, §3.3.2, Theorem 7] confine the eccentricities through the conservation of angular momentum in a sharp Nekhoroshev estimate for the planar three-body problem, which is local near a periodic orbit of the unperturbed system [3, §3.3.3]. Pinzari's construction of Kolmogorov tori in the planetary problem uses an averaging of the fast angles with an exponentially small remainder [9, Proposition 5.1]. In the other direction, Clarke, Féjoz and Guardia construct orbits of the problem with three planets along which the semi-major axis of the outer planet undergoes large variations [4, Theorem 1]; along these orbits the conservation of angular momentum does not confine the secular variables [4, §1.3].

What we add is the form of the statement: the hypothesis is exactly the strong AMD-stability condition at the initial actions, with only a fixed margin; the constants are uniform over the whole corresponding sublevel set, including large eccentricities and, in space, arbitrary inclinations; the nonresonant drift is $O(\sqrt{\mu})$ with time exponent $1/(2(\tau+1))$; and the conclusion is the ordering of the orbital annuli, for canonical and for osculating elements. To our knowledge, the statement in this sharp form has not appeared. We do not present the normal form as new.

Formal verification. Theorems A–C and Corollary D have been formalized and checked in the proof assistant Lean 4 with the Mathlib library [1], with the hypotheses, quantifier order and conclusions stated in Section 2. A formal proof of Theorem E is in preparation in the same repository. Complete proofs, with all details omitted here, are written out in [2]. Appendix A describes what is verified, the correspondence between the statements and the formal declarations, and what is not covered by the formal check.

Organization. Section 2 sets up the model and states the results. Section 3 proves the barrier and the identities for the angular momentum and the AMD. Section 4 constructs the analytic domain and the normal form and proves Theorem B, and Section 5 treats the spatial case and Corollary D. Section 6 proves Theorem E. Section 7 discusses limitations and open questions, and Appendix A the formalization.

2 Setting and results

2.1 The model

We use units in which the gravitational constant and the mass of the star are equal to one. Fix $n \geq 2$ and $\nu_1, \dots, \nu_n > 0$; the planets have masses $m_i = \mu\nu_i$, where $\mu > 0$ is small. The space dimension is $d \in \{2, 3\}$. Let $q_i \in \mathbb{R}^d$ be the heliocentric position of the i-th planet and $p_i \in \mathbb{R}^d$ its momentum, scaled so that the physical momentum is μp_i. With the symplectic form $\sum_i dp_i \wedge dq_i$, the motion is governed by

$$H_\mu = \sum_{i=1}^{n} \Big(\frac{|p_i|^2}{2\nu_i} - \frac{\nu_i}{|q_i|} \Big) + \mu f \,, \qquad f = \frac{1}{2} \Big| \sum_{i=1}^{n} p_i \Big|^2 - \sum_{1 \leq i < j \leq n} \frac{\nu_i \nu_j}{|q_i - q_j|} \,, \tag{2.1}$$

on the set where $q_i \neq 0$ and $q_i \neq q_j$ for $i \neq j$, in physical time. This is the translation reduction of the Newtonian $(1+n)$-body problem at zero total momentum, divided by μ. A solution of (2.1) determines a solution of Newton's equations with masses $(1, \mu\nu_1, \dots, \mu\nu_n)$ in barycentric coordinates, its *barycentric reconstruction* (Lemma 3.1).

Elements and the AMD. For each planet let $v_i = p_i/\nu_i$, $E_i = |v_i|^2/2 - 1/|q_i|$ and $\mathbf{e}_i = (|v_i|^2 - 1/|q_i|)q_i - (q_i \cdot v_i)v_i$ be the canonical velocity, the specific energy and the eccentricity vector. When $E_i < 0$, the semi-major axis, the Delaunay action, the eccentricity, the angular momenta and the (normalized) angular momentum deficit are

$$a_i = -\frac{1}{2E_i} \,, \quad \Lambda_i = \nu_i \sqrt{a_i} \,, \quad e_i = |\mathbf{e}_i| \,, \quad L_i = q_i \times p_i \,, \quad L = \sum_{i=1}^{n} L_i \,, \quad C = \sum_{i=1}^{n} \Lambda_i - |L| \,. \tag{2.2}$$

In the plane, $q \times p = q^1 p^2 - q^2 p^1$ is a scalar. If $E_i < 0$, then $e_i \leq 1$ and $r_i^- \leq |q_i| \leq r_i^+$, where $r_i^\pm = a_i(1 \pm e_i)$ are the perihelion and aphelion radii (Lemma 3.2). The physical angular momentum and AMD are μL and μC, so a normalized margin δ is a physical margin $\mu\delta$.

The chamber and the thresholds. Ordered axes correspond to actions in the *chamber*

$$\mathcal{Q} = \big\{ \Lambda \in \mathbb{R}^n : \ \Lambda_i > 0, \ a_1(\Lambda) < \cdots < a_n(\Lambda) \big\} \,, \qquad a_i(\Lambda) = (\Lambda_i/\nu_i)^2 \,. \tag{2.3}$$

Let $D(u) = 1 - \sqrt{1-u^2}$ for $u \in [0,1]$. For $\Lambda \in \mathcal{Q}$ and $i < j$, define

$$\begin{aligned} B_{ij}(\Lambda) &= \min \big\{ \Lambda_i D(u) + \Lambda_j D(v) : \ u, v \in [0,1], \ a_i(\Lambda)(1+u) \geq a_j(\Lambda)(1-v) \big\} \,, \\ B_{\mathrm{cross}}(\Lambda) &= \min_{1 \leq i < n} B_{i,i+1}(\Lambda) \,, \qquad B_\star(\Lambda) = \min_{1 \leq i \leq n} \Lambda_i \,, \qquad B(\Lambda) = \min \big(B_{\mathrm{cross}}(\Lambda), B_\star(\Lambda) \big) \,. \end{aligned} \tag{2.4}$$

The minima are attained and B is positive and continuous on $\mathcal{Q}$ (Lemma 3.4). Thus $B_{ij}(\Lambda)$ is the least eccentricity deficit for which the annuli of planets i and j touch, and Λ_i is the deficit of a radial orbit. Since $C \geq \sum_i \Lambda_i D(e_i)$ (Lemma 3.3), the inequality $C < B(\Lambda)$ excludes contact of adjacent annuli and $e_i = 1$. After division by Λ_j, B_{ij} is the critical AMD of [6, §3.2], and $C < B(\Lambda)$ is the strong AMD-stability of [6, §5.3] (Lemma 3.4).

Frequencies and data. The Kepler Hamiltonian and its mean motions are

$$h(\Lambda) = -\sum_{i=1}^{n} \frac{\nu_i^3}{2\Lambda_i^2} \,, \qquad \omega_i(\Lambda) = \partial_{\Lambda_i} h(\Lambda) = \frac{\nu_i^3}{\Lambda_i^3} = a_i(\Lambda)^{-3/2} \,. \tag{2.5}$$

For $\gamma > 0$ and $\tau > n-1$, a vector Λ^0 is (γ,τ)-*Diophantine* if

$$|k \cdot \omega(\Lambda^0)| \geq \gamma\, |k|_1^{-\tau} \qquad \text{for every } k \in \mathbb{Z}^n \setminus \{0\}\,, \qquad |k|_1 = \sum_{i=1}^{n} |k_i|\,. \tag{2.6}$$

Throughout, $\mathcal{A} \subset \{a \in \mathbb{R}^n : 0 < a_1 < \cdots < a_n\}$ is compact and $\delta > 0$. An initial state is *admissible* if $q_i(0) \neq 0$ and $E_i(0) < 0$ for every i, and

$$a(0) \in \mathcal{A}\,, \qquad C(0) \leq B(\Lambda(0)) - \delta\,. \tag{2.7}$$

Here $a(0) \in \mathcal{A}$ implies $\Lambda(0) \in \mathcal{Q}$, so $B(\Lambda(0))$ is defined. For $T_0, c > 0$ we put

$$T_\mu = T_0 \exp\left(c\, \mu^{-\frac{1}{2(\tau+1)}}\right), \tag{2.8}$$

and we write $\|x\|_\infty = \max_i |x_i|$ for vectors of actions or axes.

Osculating elements. The actual heliocentric velocity is $V_i = \dot{q}_i = p_i/\nu_i + \mu \sum_j p_j$, not v_i. With $k_i = 1 + \mu\nu_i$, the velocity-osculating elements are

$$E_i^{\mathrm{osc}} = \frac{|V_i|^2}{2} - \frac{k_i}{|q_i|}\,, \quad a_i^{\mathrm{osc}} = -\frac{k_i}{2E_i^{\mathrm{osc}}}\,, \quad \mathbf{e}_i^{\mathrm{osc}} = \frac{1}{k_i}\Big(\Big(|V_i|^2 - \frac{k_i}{|q_i|}\Big) q_i - (q_i \cdot V_i)\, V_i\Big)\,, \quad e_i^{\mathrm{osc}} = |\mathbf{e}_i^{\mathrm{osc}}|\,,$$

and $r_i^{\mathrm{osc},\pm} = a_i^{\mathrm{osc}}(1 \pm e_i^{\mathrm{osc}})$. When $E_i^{\mathrm{osc}} < 0$ they describe the Kepler ellipse through q_i tangent to V_i, the orbit computed from the actual position and velocity (Lemma 4.9).

2.2 Results

Theorem A (Sharp barrier)**.** *Let $d = 2$, $n \geq 2$, $\nu \in (0,\infty)^n$, let $U \subset \mathcal{Q}$ be compact and $m > 0$. There exist $d_\star, d_{\mathrm{pair}} > 0$ and a compact set K of states with $q_i \neq 0$, $q_i \neq q_j$ $(i \neq j)$ and $E_i < 0$ for all i, such that:*

(i) if $\Lambda \in U$, $e \in [0,1]^n$ and $\sum_i \Lambda_i D(e_i) \leq B(\Lambda) - m$, then $a_i(\Lambda)(1-e_i) \geq d_\star$ for $1 \leq i \leq n$ and $a_{i+1}(\Lambda)(1-e_{i+1}) - a_i(\Lambda)(1+e_i) \geq d_{\mathrm{pair}}$ for $1 \leq i < n$;

(ii) every state with $q_i \neq 0$ and $E_i < 0$ for all i, $\Lambda \in U$ and $C \leq B(\Lambda) - m$ belongs to K and satisfies $r_i^- \geq d_\star$, $r_{l+1}^- - r_l^+ \geq d_{\mathrm{pair}}$, $|q_i| \geq d_\star$, $|q_i - q_j| \geq d_{\mathrm{pair}}$, $|L_i| \geq m$ and $L_i L > 0$, for all $i \neq j$ and $1 \leq l < n$.

Theorem B (Planar exponential collision safety)**.** *Let $d = 2$, $n \geq 2$, $\nu \in (0,\infty)^n$, $\mathcal{A}$ and δ as above, $\gamma > 0$ and $\tau > n-1$. There exist positive constants $\mu_0, c, T_0, A_\Lambda, A_a, d_\star, d_{\mathrm{pair}}, A_{\mathrm{osc}}, A_{\mathrm{el}}$, depending only on $(n, \nu, \mathcal{A}, \delta, \gamma, \tau)$ and satisfying $A_{\mathrm{osc}}\mu_0 \leq d_\star/2$ and $2A_{\mathrm{osc}}\mu_0 \leq d_{\mathrm{pair}}/2$, such that the following holds for every $\mu \in (0, \mu_0]$ and every admissible initial state with (γ,τ)-Diophantine $\Lambda(0)$. Let $T = T_\mu$.*

(i) Well-posedness. *There is a solution of (2.1) on $[-T,T]$ with the given initial state. Every solution with the same initial state on an open interval containing 0 agrees with it on the common part of $[-T,T]$. Its barycentric reconstruction solves Newton's equations with masses $(1, \mu\nu_1, \ldots, \mu\nu_n)$ on $[-T,T]$.*

Moreover, for every $t \in [-T,T]$, every $i \neq j$ and every $1 \leq l < n$, with all elements evaluated at time t:

(ii) AMD margin. *$q_i \neq 0$, $E_i < 0$, $\Lambda(t) \in \mathcal{Q}$ and $C(t) \leq B(\Lambda(t)) - \delta/2$.*

(iii) Drift. $\|\Lambda(t)-\Lambda(0)\|_\infty \le A_\Lambda\sqrt{\mu}$, $\|a(t)-a(0)\|_\infty \le A_a\sqrt{\mu}$ *and* $|C(t)-C(0)| \le nA_\Lambda\sqrt{\mu}$.

(iv) Separation. $r_i^- \ge d_\star$, $r_{l+1}^- - r_l^+ \ge d_{\mathrm{pair}}$, $|q_i| \ge d_\star$ *and* $|q_i - q_j| \ge d_{\mathrm{pair}}$.

(v) Osculating orbits. $E_i^{\mathrm{osc}} < 0$, $a_i^{\mathrm{osc}} > 0$, $e_i^{\mathrm{osc}} < 1$, $r_i^{\mathrm{osc},-} \le |q_i| \le r_i^{\mathrm{osc},+}$, *and*

$$|a_i^{\mathrm{osc}} - a_i| + |\mathbf{e}_i^{\mathrm{osc}} - \mathbf{e}_i| \le A_{\mathrm{el}}\mu\,, \quad |r_i^{\mathrm{osc},\pm} - r_i^{\pm}| \le A_{\mathrm{osc}}\mu\,, \quad r_i^{\mathrm{osc},-} \ge \frac{d_\star}{2}\,, \quad r_{l+1}^{\mathrm{osc},-} - r_l^{\mathrm{osc},+} \ge \frac{d_{\mathrm{pair}}}{2}\,.$$

Theorem C (Spatial exponential collision safety)**.** *For $d = 3$, Theorem B holds verbatim, with the same data and quantifiers and the conclusions* (i)–(v)*, where $|\cdot|$ is the Euclidean norm on $\mathbb{R}^3$ and L is the vector total angular momentum.*

Corollary D (Exponent $1/(2n+1)$)**.** *Theorems B and C hold with $\tau = n - 1/2$, and then $T_\mu = T_0 \exp(c\mu^{-1/(2n+1)})$.*

Theorem E (Collision safety without arithmetic hypothesis)**.** *Let $d \in \{2,3\}$, $n \ge 2$, $\nu \in (0,\infty)^n$, $\mathcal{A}$ and δ as above, and put $\mathfrak{a}_n = 2/(6n^2-3n+14)$ and $\mathfrak{b}_n = 3/(6n^2-3n+14)$. There exist positive constants $\mu_0, c, T_0, A_\Lambda, A_a, d_\star, d_{\mathrm{pair}}, A_{\mathrm{osc}}, A_{\mathrm{el}}$, depending only on $(n, \nu, \mathcal{A}, \delta)$ and satisfying $A_{\mathrm{osc}}\mu_0 \le d_\star/2$ and $2A_{\mathrm{osc}}\mu_0 \le d_{\mathrm{pair}}/2$, such that for every $\mu \in (0, \mu_0]$ and every admissible initial state, with $T = T_0 \exp(c\mu^{-\mathfrak{a}_n})$, the conclusions* (i), (ii), (iv) *and* (v) *of Theorem B hold, and* (iii) *holds with $\sqrt{\mu}$ replaced by $\mu^{\mathfrak{b}_n}$.*

2.3 Comments

Constants. All constants are chosen from the data, before μ and the initial state, and are uniform over all phases, eccentricities, inclinations and orientations allowed by (2.7). The bounds on $A_{\mathrm{osc}}\mu_0$ make the osculating separations in (v) consequences of (iv). The masses are not normalized; the proofs may assume $\max_i \nu_i = 1$, since with $M = \max_i \nu_i$ the substitution $(\mu, \nu, p) \mapsto (M\mu, \nu/M, p/M)$ preserves the masses, the motion, the axes and the frequencies, and divides Λ, C and B by M.

What is not claimed. The theorems concern the limit $\mu \to 0$ with fixed δ, $\mathcal{A}$ and mass ratios, with non-explicit constants; nothing is asserted about any actual planetary system. No secular stability is assumed or proved: the eccentricities and inclinations may evolve chaotically inside $\{C \le B(\Lambda) - \delta/2\}$.

No orbit crossing. Conclusion (iv) says that the entire annuli $\{r_i^- \le |x| \le r_i^+\}$ stay strictly ordered, at distance at least d_{pair} from each other and $d_\star$ from the star. This is stronger than separation of the current positions. Conclusion (v) gives the same for the ellipses determined by the actual positions and velocities, with half the distances.

Theorems B and E. Neither implies the other: Theorem B excludes resonant frequencies, while Theorem E gives the larger drift $\mu^{\mathfrak{b}_n}$ ($\mathfrak{b}_n < 1/2$) and, if $2(\tau+1)\mathfrak{a}_n < 1$, a shorter time. For $n = 2$, $\tau = 3/2$ the time exponents are $1/5$ and $1/16$.

Space. Theorem C makes no hypothesis on inclinations or on the orientation of the orbits; admissible states may be circular or coplanar, and $C < B_\star$ already forces $L \neq 0$ and $L_i \cdot L > 0$ (Lemma 3.3). The proof of Theorem A (Section 3) does not use the dimension; for $d = 3$ it gives the same conclusions with $|L_i| \ge m$ and $L_i L > 0$ replaced by $L_i \cdot L \ge m|L|$, and this version enters the proof of Theorem C.

3 The sharp barrier

This section is elementary and contains no small parameter. We show that exact conservation of the total angular momentum turns the strong condition $C \le B(\Lambda) - m$ into a bound on the eccentricities that keeps every perihelion and every pair of adjacent radial annuli uniformly apart, and that the threshold B is exactly the least deficit at which this fails. Throughout, $d \in \{2, 3\}$, and in the plane $q \times p$ is the oriented scalar determinant.

Lemma 3.1 (Normalization)**.** *For every $d \in \{2, 3\}$ and all $\nu_i, \mu > 0$, the Hamiltonian (2.1) is the translation reduction, at zero total momentum, of the Newtonian $(1+n)$-body Hamiltonian with masses $(1, \mu\nu_1, \dots, \mu\nu_n)$, divided by μ, with unchanged time. Its solutions reconstruct Newtonian solutions with these masses.*

Proof. In barycentric coordinates put $P_i = \mu p_i$ and $P_0 = -\sum_i P_i$. With $q_i = x_i - x_0$ we have $\sum_{i=0}^{n} P_i \cdot dx_i = \sum_{i=1}^{n} P_i \cdot dq_i$; the kinetic energy becomes $\sum_i |P_i|^2/(2\mu\nu_i) + 1/2|\sum_i P_i|^2$ and the potential $-\sum_i \mu\nu_i/|q_i| - \sum_{i<j} \mu^2\nu_i\nu_j/|q_i - q_j|$. The symplectic form and the Hamiltonian both acquire the factor μ, which cancels and gives (2.1). Conversely, with $M = 1 + \mu\sum_i \nu_i$ put $x_0 = -\mu\sum_i \nu_i q_i/M$ and $x_i = x_0 + q_i$. Hamilton's equation $\dot q_i = p_i/\nu_i + \mu\sum_j p_j$ gives $\dot x_0 = -\mu\sum_j p_j$ and $\dot x_i = p_i/\nu_i$, and the equations for $\dot p_i$ are Newton's equations after multiplying the planet momenta by μ. □

Lemma 3.2 (Orbital identities)**.** *Let $\nu > 0$, $q \neq 0$, $v = p/\nu$ and $E = |v|^2/2 - 1/|q| < 0$, and define $a, \Lambda, \mathbf{e}, e$ as in (2.2). Then*

$$e^2 = 1 + 2E|q \times v|^2 = 1 - \frac{|q \times v|^2}{a} \le 1, \qquad |q \times p| = \Lambda\sqrt{1 - e^2}, \tag{3.1}$$

$$a(1 - e) \le |q| \le a(1 + e), \qquad |p|^2 = \nu^2\Big(\frac{2}{|q|} - \frac{1}{a}\Big), \tag{3.2}$$

including the radial case $e = 1$.

Proof. Expanding $|(|v|^2 - 1/|q|)q - (q \cdot v)v|^2$ and using $|q \times v|^2 = |q|^2|v|^2 - (q \cdot v)^2$ gives the first identity in (3.1); substitute $2E = -1/a$ for the rest. With $r = |q|$, the energy identity and the radial/tangential decomposition of v give $(q \cdot v)^2 = 2r - r^2/a - |q \times v|^2 \ge 0$. Inserting $|q \times v|^2 = a(1 - e^2)$ and completing the square gives $(r - a)^2 \le a^2e^2$. The last identity in (3.2) is the definition of E. □

Lemma 3.3 (Angular momentum and the AMD identity)**.** *Along every solution of (2.1) on an interval, L is constant. In the negative-energy region $C \ge \sum_i \Lambda_i D(e_i) \ge 0$. If $C < B_\star(\Lambda)$, then $L \neq 0$, the projections $H_i = L_i \cdot L/|L|$ are positive, and*

$$C(t) - C(0) = \sum_i \big(\Lambda_i(t) - \Lambda_i(0)\big), \qquad C = \sum_i (\Lambda_i - H_i). \tag{3.3}$$

If $C \le B_\star(\Lambda) - m$ with $m > 0$, then $H_i \ge m$ for every i. In the plane, after choosing the orientation with $L > 0$, every L_i is positive and $H_i = L_i$.

Proof. Differentiate $\sum_i q_i \times p_i$. The kinetic contributions are $(p_i/\nu_i) \times p_i = 0$ and $\mu(\sum_j p_j) \times (\sum_i p_i) = 0$; the central forces have zero torque, and the two forces of a pair contribute a multiple of $(q_i - q_j) \times (q_i - q_j) = 0$. Hence $\dot L = 0$. By (3.1), $|L| \le \sum_i |L_i| = \sum_i \Lambda_i\sqrt{1 - e_i^2}$, which gives $C \ge \sum_i \Lambda_i D(e_i) \ge 0$. If $L = 0$, then $C = \sum_i \Lambda_i \ge B_\star$. Otherwise $|L| = \sum_i H_i$ gives the second identity in (3.3), and constancy of $|L|$ gives the first. Since each $\Lambda_i - H_i \ge \Lambda_i - |L_i| \ge 0$, we get $H_i \ge \Lambda_i - C$, which is positive if $C < B_\star$ and at least m if $C \le B_\star - m$. □

Lemma 3.4 (The thresholds)**.** *For all $\nu_i > 0$ and $\Lambda \in \mathcal{Q}$, every minimum in (2.4) is attained and positive, and B_{ij} and B are continuous on $\mathcal{Q}$. The minimum defining B_{ij} is attained at radial contact $a_i(1+u) = a_j(1-v)$, and*

$$B_{ij}(\Lambda) < \Lambda_j, \qquad B(\Lambda) = \min\big(B_{\mathrm{cross}}(\Lambda), \Lambda_1\big). \tag{3.4}$$

With $\alpha = a_i/a_j$, the quotient $B_{ij}(\Lambda)/\Lambda_j$ is the relative critical AMD $C_c(\alpha, \nu_i/\nu_j)$ of Laskar and Petit [6, §§3.1–3.2].

Proof. Fix $i < j$, so $0 < \alpha < 1$. The feasible set $\{(u,v) \in [0,1]^2 : \alpha u + v \geq 1 - \alpha\}$ is compact and contains $(0, 1-\alpha)$, and the objective $\Lambda_i D(u) + \Lambda_j D(v)$ is continuous, so the minimum is attained. If (u,v) is feasible, then $t = (1-\alpha)/(\alpha u + v) \leq 1$, the point (tu, tv) lies on the contact line $\alpha u + v = 1 - \alpha$, and since D is increasing the cost does not increase. Hence the minimum is attained at contact, which we parametrize over a fixed interval:

$$u = \bar{u}_\alpha t, \qquad v = 1 - \alpha - \alpha \bar{u}_\alpha t, \qquad t \in [0,1], \qquad \bar{u}_\alpha = \min\Big(1, \frac{1-\alpha}{\alpha}\Big). \tag{3.5}$$

The resulting cost $F(\Lambda, t)$ is jointly continuous, hence uniformly continuous on $K \times [0,1]$ for every compact $K \subset \mathcal{Q}$, and $|\min_t F(\Lambda, t) - \min_t F(\Lambda', t)| \leq \sup_t |F(\Lambda, t) - F(\Lambda', t)|$ proves continuity of B_{ij}. A zero cost forces $u = v = 0$, which is not feasible. Finite minima of positive continuous functions are positive and continuous. The competitor $(0, 1-\alpha)$ gives $B_{ij} \leq \Lambda_j D(1-\alpha) < \Lambda_j$. In particular $\Lambda_i > B_{i-1,i} \geq B_{\mathrm{cross}}$ for $i \geq 2$, which gives the second identity in (3.4).

Finally $\Lambda_i/\Lambda_j = (\nu_i/\nu_j)\sqrt{\alpha}$, so B_{ij}/Λ_j is the minimum of $(\nu_i/\nu_j)\sqrt{\alpha}\, D(u) + D(v)$ on the contact line $\alpha u + v = 1 - \alpha$, $u, v \in [0,1]$. This is exactly the minimization of the relative AMD under the collision condition that defines $C_c(\alpha, \gamma)$ in [6, §§3.1–3.2], with mass ratio $\gamma = \nu_i/\nu_j$. □

The threshold is geometrically sharp: the minimizers of B_{ij} are nondegenerate eccentricities, two coplanar prograde ellipses with these eccentricities, opposite perihelion directions, and the inner aphelion and outer perihelion on one ray touch with combined AMD B_{ij}, and a radial orbit has deficit Λ_i, approached by prograde ellipses as $e_i \uparrow 1$. This is sharpness of the geometric condition only, not a claim that motion above the threshold must collide.

Lemma 3.5 (Uniform barrier)**.** *Let $U \subset \mathcal{Q}$ be compact and $m > 0$. There exist $d_\star, d_{\mathrm{pair}} > 0$, depending only on (n, ν, U, m), such that for every $\Lambda \in U$ and $e \in [0,1]^n$ with $\sum_i \Lambda_i D(e_i) \leq B(\Lambda) - m$,*

$$a_i(\Lambda)(1 - e_i) \geq d_\star, \qquad a_{i+1}(\Lambda)(1 - e_{i+1}) - a_i(\Lambda)(1 + e_i) \geq d_{\mathrm{pair}}. \tag{3.6}$$

Every state with $q_i \neq 0$, $E_i < 0$, $\Lambda \in U$ and $C \leq B(\Lambda) - m$ satisfies (3.6), $|q_i| \geq d_\star$ and $|q_i - q_j| \geq d_{\mathrm{pair}}$ for $i \neq j$, and $H_i \geq m$; these states form a compact subset of the region $\{q_i \neq 0,\ q_i \neq q_j,\ E_i < 0\}$.

Proof. Let $K_m = \{(\Lambda, e) \in U \times [0,1]^n : \sum_i \Lambda_i D(e_i) \leq B(\Lambda) - m\}$, which is compact by Lemma 3.4. If an adjacent separation in (3.6) were nonpositive at a point of K_m, then (e_i, e_{i+1}) would be feasible for $B_{i,i+1}$ and $\sum_k \Lambda_k D(e_k) \geq B_{i,i+1}(\Lambda) \geq B(\Lambda)$, a contradiction. The separations are continuous, so if $K_m \neq \emptyset$ their minimum over K_m is a positive number d_{pair} (if $K_m = \emptyset$ all assertions are vacuous). Put

$$\Lambda_{\max} = \max_{\Lambda \in U,\, i} \Lambda_i, \qquad a_{\min} = \min_{\Lambda \in U,\, i} a_i(\Lambda), \qquad a_{\max} = \max_{\Lambda \in U,\, i} a_i(\Lambda), \qquad d_\star = \frac{a_{\min} m^2}{2\Lambda_{\max}^2}. \tag{3.7}$$

Since $B \le B_\star \le \Lambda_i$, the deficit bound gives $\Lambda_i D(e_i) \le \Lambda_i - m$, that is $\sqrt{1-e_i^2} \ge m/\Lambda_{\max}$, and $1 - e_i \ge (1-e_i^2)/2$ gives the first bound in (3.6).

For a physical state, Lemma 3.3 gives $\sum_i \Lambda_i D(e_i) \le C \le B(\Lambda) - m$, so (3.6) holds, and $H_i \ge m$ because $B \le B_\star$. By (3.2), $d_\star \le |q_i| \le 2a_{\max}$, and summing the adjacent separations between the annuli gives $|q_i - q_j| \ge |q_j| - |q_i| \ge d_{\text{pair}}$ for $i < j$. Next, (3.2) gives $|p_i| \le \nu_i\sqrt{2/d_\star}$, and $E_i \le -1/(2a_{\max})$. The closed bounded set defined by these inequalities lies in the nonsingular region, and the conditions $\Lambda \in U$, $C \le B(\Lambda) - m$ are closed there; so the set of such states is compact. □

Theorem A is Lemma 3.5: in the plane, with the orientation $L > 0$ of Lemma 3.3, $H_i = L_i = |L_i| \ge m$.

The next lemma converts the conservation law into a margin that moves with the actions. It is the only place where the drift of the actions is compared with the threshold.

Lemma 3.6 (Moving threshold and continuation)**.** *Let $\mathcal{L} \subset \mathcal{Q}$ be compact and $\delta > 0$. There exist $r_g > 0$ and a compact set $U \subset \mathcal{Q}$ containing the closed $\|\cdot\|_\infty$-neighbourhood of radius $4r_g$ of $\mathcal{L}$ such that the modulus*

$$\Omega_B(r) = \sup\{|B(\Lambda) - B(\Lambda')| : \Lambda, \Lambda' \in U,\ \|\Lambda - \Lambda'\|_\infty \le r\}$$

satisfies $\Omega_B(r) \to 0$ as $r \downarrow 0$ and $\Omega_B(4r_g) \le \delta/16$. Let $0 < r \le 4r_g$. If a solution of (2.1) has $\Lambda^0 = \Lambda(0) \in \mathcal{L}$, $C(0) \le B(\Lambda^0) - \delta$, and $\|\Lambda(t) - \Lambda^0\|_\infty \le r$ while $E_i(t) < 0$, then

$$C(t) \le B(\Lambda(t)) - \delta + nr + \Omega_B(r). \tag{3.8}$$

If $nr + \Omega_B(r) \le \delta/2$, the solution cannot reach the end of its maximal interval of existence at a finite time while these action and ellipticity conditions hold, forwards or backwards in time.

Proof. Take any $r_0 > 0$ with the closed $2r_0$-neighbourhood U of $\mathcal{L}$ inside $\mathcal{Q}$. By Lemma 3.4, B is uniformly continuous on U, so $\Omega_B(r) \to 0$, and we choose $4r_g \le r_0$ with $\Omega_B(4r_g) \le \delta/16$. Lemma 3.3 gives $C(t) = C(0) + \sum_i(\Lambda_i(t) - \Lambda_i^0) \le B(\Lambda^0) - \delta + nr$, and $B(\Lambda^0) \le B(\Lambda(t)) + \Omega_B(r)$ gives (3.8). If $nr + \Omega_B(r) \le \delta/2$, then $C(t) \le B(\Lambda(t)) - \delta/2$, and Lemma 3.5 with $m = \delta/2$ keeps the state in a fixed compact subset of the domain of the smooth vector field of (2.1). A maximal solution of a locally Lipschitz autonomous equation leaves every compact subset of its domain at a finite endpoint, so such an endpoint cannot occur while the conditions hold. For backward time replace the vector field by its negative. □

A single compact set U and a single modulus Ω_B serve for the rest of the paper: they are fixed by Lemma 3.6 with $\mathcal{L}$ replaced by the set $\mathcal{L}_\delta$ of Section 4.

4 The planar theorem

By Section 3, Theorem B reduces to the actions: if $\Lambda(t)$ stays within $O(\sqrt{\mu})$ of Λ^0 for $|t| \le T_\mu$, then (3.8) keeps the AMD $\delta/2$ below the moving threshold, and Lemma 3.5 gives every separation. We control the actions by averaging over the fast angles, with three features.

First, we average about the *linear* Hamiltonian $h_0 = h(\Lambda^0) + \omega(\Lambda^0)\cdot(\Lambda - \Lambda^0)$ with the initial Kepler frequencies; the Kepler nonlinearity $h - h_0$ is part of the perturbation on an action tube of radius r. Its vector field has size r, that of μf has size μ/r in the action directions, and $r = \sqrt{\mu}$ balances the two. Second, the secular variables $z = (x_i, y_i)_i$ are carried along as canonical variables,

with no assumption on their dynamics; the averaged Hamiltonian is never analysed. Third, the slow domain is the entire ball of secular states allowed by the AMD margin; all its configurations are collision free with a fixed margin, so it is one analytic domain with fixed complex widths. Since the AMD equals $1/2|z|^2$, the orbit cannot leave the ball before its actions drift or its margin is lost, and a first-exit argument closes the bootstrap.

4.1 The chart and its analytic domain

We use the regular Poincaré variables. For one body with parameter $\nu > 0$, given $\Lambda > 0$ and $(x, y) \in \mathbb{R}^2$ with $\Gamma = (x^2 + y^2)/2 < \Lambda$, put

$$a = (\Lambda/\nu)^2, \qquad \beta = 1 - \frac{\Gamma}{\Lambda}, \qquad \varrho = \sqrt{\frac{1 - \Gamma/(2\Lambda)}{\Lambda}}, \qquad k = \varrho x, \qquad l = -\varrho y,$$

so that $k^2 + l^2 = 1 - \beta^2 < 1$. For $\lambda \in \mathbb{R}$ the eccentric longitude F is the unique solution of

$$F - k \sin F + l \cos F = \lambda, \tag{4.1}$$

and with $c(F) = (\cos F, \sin F)$, $c'(F) = (-\sin F, \cos F)$ and $D_F = 1 - k \cos F - l \sin F$ we set

$$\mathbf{M} = \begin{pmatrix} 1 - \frac{l^2}{1+\beta} & \frac{kl}{1+\beta} \\ \frac{kl}{1+\beta} & 1 - \frac{k^2}{1+\beta} \end{pmatrix}, \qquad q = a\big(\mathbf{M}c(F) - (k, l)\big), \qquad p = \frac{\nu}{\sqrt{a}} \frac{\mathbf{M}c'(F)}{D_F}. \tag{4.2}$$

Lemma 4.1 (Regular planar chart)**.** *For every $\nu > 0$, the map $(\Lambda, \lambda, x, y) \mapsto (q, p)$ of (4.1)–(4.2), with λ read modulo 2π, is a real analytic diffeomorphism from $\{\Lambda > 0,\ \lambda \in \mathbb{T},\ (x, y) \in \mathbb{R}^2,\ \Gamma < \Lambda\}$ onto the planar states with $q \neq 0$, $E < 0$ and $q \times p > 0$. It carries $dp \wedge dq$ to $d\Lambda \wedge d\lambda + dx \wedge dy$; in it the Kepler energy $|p|^2/(2\nu) - \nu/|q|$ is $-\nu^3/(2\Lambda^2)$, the angular momentum is $\Lambda - \Gamma$, and the eccentricity vector is (k, l). It is regular at circular orbits.*

Sketch. The left side of (4.1) has derivative $D_F \geq 1 - \sqrt{k^2 + l^2} > 0$ and increases by 2π per period, so F exists, is unique and is analytic. Where $e = \sqrt{k^2 + l^2} > 0$, write $(k, l) = e(\cos \varpi, \sin \varpi)$ and $F = \psi + \varpi$. In the frame rotated by ϖ, (4.2) is the classical parametrization by the eccentric anomaly ψ, with $\beta = \sqrt{1 - e^2}$; it gives the energy, the angular momentum $G = \Lambda\beta = \Lambda - \Gamma$ and the eccentricity vector, also at $e = 0$ by continuity. With $\ell = \psi - e \sin \psi$, so that $\lambda = \ell + \varpi$, differentiation and the contribution $G\, d\varpi$ of the rotation give

$$p \cdot dq = \Lambda\, d\ell + G\, d\varpi + d(2\Lambda e \sin \psi) = \Lambda\, d\lambda + \tfrac{1}{2}(x\, dy - y\, dx) + d(2\Lambda e \sin \psi),$$

because $(x, y) = \sqrt{2\Gamma}(\cos \varpi, -\sin \varpi)$ gives $\Gamma\, d(-\varpi) = \frac{1}{2}(x\, dy - y\, dx)$. Exterior differentiation proves the symplectic identity for $e > 0$, and analyticity extends it to $e = 0$. The inverse recovers Λ from the energy, Γ from the angular momentum, (k, l) as the eccentricity vector, $c(F) = \mathbf{M}^{-1}(q/a + (k, l))$, and $e^{i\lambda} = e^{iF} \exp(-i(k \sin F - l \cos F))$, analytically and without a perihelion angle; see [2]. □

For n planets we use the product chart, with $z = (x_i, y_i)_{i \leq n} \in \mathbb{R}^{2n}$. It covers the states with $q_i \neq 0$, $E_i < 0$ and $L_i > 0$ for all i; on them $H_\mu = h(\Lambda) + \mu f(\Lambda, \lambda, z)$ exactly, f being the pulled-back perturbation of (2.1), and if moreover $L > 0$,

$$C = \sum_i \Lambda_i - \sum_i L_i = \sum_i \Gamma_i = \tfrac{1}{2}|z|^2. \tag{4.3}$$

Let $\mathcal{L} = \{(\nu_i\sqrt{a_i})_i : a \in \mathcal{A}\} \subset \mathcal{Q}$ and $\mathcal{L}_\delta = \{\Lambda^0 \in \mathcal{L} : B(\Lambda^0) \geq \delta\}$. Since $C \geq 0$, every initial state satisfying (2.7) has $\Lambda^0 \in \mathcal{L}_\delta$. We fix r_g, U, Ω_B by Lemma 3.6 applied to $\mathcal{L}_\delta$. For complex arguments, $|z|^2$ and Γ_i are bilinear sums of squares, and distances are in the maximum norm.

Lemma 4.2 (Analytic domain of the whole sublevel ball)**.** *There are $r_* \in (0, 4r_g]$ and $s, b, M > 0$, depending only on $(n, \nu, \mathcal{A}, \delta)$, such that for every $\Lambda^0 \in \mathcal{L}_\delta$, with*

$$\mathcal{Z}_0(\Lambda^0) = \left\{z \in \mathbb{R}^{2n} : \tfrac{1}{2}|z|^2 \leq B(\Lambda^0) - \delta/4\right\}, \tag{4.4}$$

the product chart and f are holomorphic, 2π-periodic in λ and real on real points in the complex domain

$$\|\Lambda - \Lambda^0\|_\infty < r_*, \qquad \|\mathrm{Im}\,\lambda\|_\infty < 4s, \qquad \mathrm{dist}_\infty\left(z, \mathcal{Z}_0(\Lambda^0)\right) < 4b, \tag{4.5}$$

and on a slightly smaller domain of the same form the chart, the momenta, f and their first derivatives are bounded by M. Every denominator $|q_i|$, $|q_i - q_j|$ of f has a single holomorphic branch there, with modulus bounded below by a positive constant depending only on $(n, \nu, \mathcal{A}, \delta)$.

Proof. We may assume $\mathcal{L}_\delta \neq \emptyset$; it is compact.

Step 1: every real point of the ball is safe. Let $\Lambda^0 \in \mathcal{L}_\delta$, Λ real with $\|\Lambda - \Lambda^0\|_\infty \leq 4r_g$ (so $\Lambda \in U$), and $z \in \mathcal{Z}_0(\Lambda^0)$. As $\Omega_B(4r_g) \leq \delta/16$,

$$\sum_i \Gamma_i = \tfrac{1}{2}|z|^2 \leq B(\Lambda^0) - \frac{\delta}{4} \leq B(\Lambda) - \frac{3\delta}{16}.$$

Since $B(\Lambda) \leq \Lambda_i$, we get $\Gamma_i \leq \Lambda_i - 3\delta/16$: the point lies in the chart domain with a fixed margin, and $e_i \leq e_* < 1$ for a fixed e_*. At every phase λ the physical state has all $L_i > 0$ and, by (4.3), $C \leq B(\Lambda) - 3\delta/16$, so Lemma 3.5 with $m = 3\delta/16$ gives $d \leq |q_i|, |q_i - q_j| \leq Q$ with fixed $d, Q > 0$. The set K' of all such (Λ, λ, z), over $\Lambda^0 \in \mathcal{L}_\delta$ and λ in a closed period cell, is compact.

Step 2: a uniform complex neighbourhood. Since $D_F \geq 1 - e_*$, the Newton map for (4.1) contracts a complex disk of fixed radius about each real root, giving a bounded holomorphic branch of F; the radicands and $1 + \beta$ stay away from zero. By the identity theorem, local extensions agree on overlaps, and a finite subcover of K' gives one complex neighbourhood on which the chart is single valued and periodic. For a denominator, if the complex vector w lies within distance η of a real w_0 with $d \leq |w_0| \leq Q$, then

$$|w \cdot w - |w_0|^2| \leq 2Q\eta + \eta^2 \leq d^2/2$$

for small fixed η, so $\mathrm{Re}(w \cdot w) \geq d^2/2$, the root with positive real part is holomorphic, and its reciprocal is at most $\sqrt{2}/d$. Cauchy's estimates bound first derivatives. Finally choose $r_* \leq 4r_g$ and s, b so small that (4.5), which lies within distance $\max(r_*, 4s, 4b)$ of K', and a slight enlargement of it lie in this neighbourhood. □

Thus the slow domain is the *entire* ball (4.4), with complex widths independent of Λ^0 and μ. The proof uses only the safety of every real point of the ball, compactness and the bilinear bound on the denominators, so it applies verbatim to any analytic chart with the identity (4.3).

4.2 Exponential averaging with evolving slow variables

The next three lemmas are standard Nekhoroshev-type estimates, in the form of Pöschel's Normal Form Lemma [10, p. 192] (see also [7]); we do not claim them as new. We use a norm on Hamiltonian vector fields, a linear unperturbed Hamiltonian, and n_s slow pairs near an arbitrary compact set;

the slow variables enter only through Cauchy estimates, which need nothing of the shape of that set. Proofs are in [2].

Fix $n \geq 1$, $n_s \geq 0$, a nonempty compact $S \subset \mathbb{R}^{2n_s}$, $\Lambda^0 \in \mathbb{R}^n$ and widths $r, s, b > 0$, use the symplectic form $d\Lambda \wedge d\lambda + \sum dx \wedge dy$, and put $\mathsf{D} = 2n + 2n_s$. For $1 \leq \rho \leq 4$ let

$$\mathcal{D}_\rho = \big\{(\Lambda, \lambda, z) : \|\Lambda - \Lambda^0\|_\infty < \rho r,\ \|\mathrm{Im}\,\lambda\|_\infty < \rho s,\ \mathrm{dist}_\infty(z, S) < \rho b\big\}. \tag{4.6}$$

For a λ-periodic holomorphic vector field V with scaled components $\widehat{V} = (V_\Lambda/r, V_\lambda/s, V_z/b)$ and Fourier coefficients $\widehat{V}_k(\Lambda, z)$, let $\|V\|_\rho = \sum_{k\in\mathbb{Z}^n} e^{\rho s|k|_1} \sup|\widehat{V}_k|_\infty$, the supremum over $\|\Lambda - \Lambda^0\|_\infty < \rho r$, $\mathrm{dist}_\infty(z, S) < \rho b$.

Lemma 4.3 (Fourier and commutator estimates)**.** *For $0 < \sigma < \rho - 1$ the norm decreases under restriction, the λ-average and its complement have norm at most one, and*

$$\|[V, W]\|_{\rho-\sigma} \leq \frac{2\mathsf{D}}{\sigma}\|V\|_\rho\|W\|_\rho. \tag{4.7}$$

If $|\widehat{V}|_\infty \leq M'$ on $\mathcal{D}_4$, then $\|V\|_3 \leq M'\big((1 + e^{-s})/(1 - e^{-s})\big)^n$.

Sketch. Contour shifts give $|\widehat{V}_k| \leq M'e^{-4s|k|_1}$. A disk of radius σr or σb about a point of the smaller domain lies in the larger one, since $\mathrm{dist}(\cdot, S)$ is 1-Lipschitz, so Cauchy's formula costs $1/\sigma$ in each of the D directions (in the angles, $s|k_j|e^{-\sigma s|k|_1} \leq 1/(e\sigma)$), which gives (4.7). □

Lemma 4.4 (Lie step)**.** *Let $|k \cdot \omega| \geq \gamma|k|_1^{-\tau}$ for $k \in \mathbb{Z}^n \setminus \{0\}$, $h_0 = \omega \cdot (\Lambda - \Lambda^0) + c_0$, and $H = h_0 + g + f$ real holomorphic on $\mathcal{D}_\rho$ with $\partial_\lambda g = 0$. Put $\mathcal{E} = \|X_f\|_\rho$, $\mathcal{G} = \|X_g\|_\rho$ and $A_\gamma = \gamma^{-1}\max\{1, (2\tau/(es))^\tau\}$. For $0 < \sigma < \rho - 1$ there is a real χ with*

$$\omega \cdot \partial_\lambda\chi = f - \langle f\rangle, \qquad \|X_\chi\|_{\rho-\sigma/2} \leq A_\gamma\sigma^{-\tau}\mathcal{E} =: u, \tag{4.8}$$

$\langle f\rangle$ the λ-average. There are $c_\mathsf{D}, B_\mathsf{D} > 0$ depending only on D such that, if $u/\sigma \leq c_\mathsf{D}$, the time-one map ϕ of X_χ is real, holomorphic, symplectic and periodic on $\mathcal{D}_{\rho-\sigma}$, moves points by at most u in scaled coordinates, and

$$H \circ \phi = h_0 + g + \langle f\rangle + f_+, \qquad \|X_{f_+}\|_{\rho-\sigma} \leq B_\mathsf{D}\frac{u}{\sigma}(\mathcal{G} + \mathcal{E}). \tag{4.9}$$

The constants do not depend on r, b, S, Λ^0.

Sketch. Put $\chi_k = f_k/(ik \cdot \omega)$ for $k \neq 0$; the divisor is constant, and $|k|_1^\tau e^{-\sigma s|k|_1/2} \leq (2\tau/(e\sigma s))^\tau$ gives (4.8). Iterating (4.7) gives $\|\mathrm{ad}^j_{X_\chi}V\|_{\rho-\eta}/j! \leq (2e\mathsf{D}u/\eta)^j\|V\|_\rho$, so the Lie series converges and represents the flow. The identity $\{h_0, \chi\} = -(f - \langle f\rangle)$ removes X_{h_0}, which is not small, exactly; the remaining terms are at least linear in χ and give (4.9). □

Lemma 4.5 (Normal form on the full slow tube)**.** *In the setting of Lemma 4.4 let $H = h_0 + F$ with $\|X_F\|_3 \leq \epsilon$, and let $N \geq 1$ be an integer. There is $q_\mathsf{D} > 0$, depending only on D, such that if*

$$A_\gamma\,\epsilon\,N^{\tau+1} \leq q_\mathsf{D}, \tag{4.10}$$

there is a real holomorphic symplectic map $\Phi : \mathcal{D}_2 \to \mathcal{D}_3$ with

$$H \circ \Phi = h_0 + Y + R, \qquad \partial_\lambda Y = 0, \qquad \|X_R\|_2 \leq \epsilon 2^{-N}, \qquad \|X_Y\|_2 \leq 2\epsilon. \tag{4.11}$$

The map Φ moves points by at most $1/16$ in scaled coordinates; its inverse is defined on $\mathcal{D}_{3/2}$, with values in $\mathcal{D}_2$ and the same bound; in particular both move actions by at most $r/16$. No condition is imposed on the slow dynamics.

Sketch. Apply Lemma 4.4 N times with $\sigma = 1/N$ on $\rho_j = 3 - j/N$, from $g_0 = 0$, $f_0 = F$. If $\mathcal{E}_j \le \epsilon 2^{-j}$ and $\mathcal{G}_j \le 2\epsilon$, then $u_j \le A_\gamma N^\tau \epsilon 2^{-j}$ and $\mathcal{E}_{j+1} \le 3B_{\mathsf{D}} A_\gamma \epsilon N^{\tau+1} \mathcal{E}_j \le \mathcal{E}_j/2$ for $q_{\mathsf{D}} = \min(c_{\mathsf{D}}, 1/(6B_{\mathsf{D}}), 1/64)$, while $\mathcal{G}_{j+1} \le \sum_{l \le j} \mathcal{E}_l \le 2\epsilon$. The total displacement $\sum_j u_j \le 2A_\gamma \epsilon N^\tau \le 1/32$ is smaller than every allotted loss, so the composition of the time-one maps is defined on $\mathcal{D}_2$ and inverted by the time-reversed flows. □

4.3 Scaling and the exponential time

We apply Lemma 4.5 in the chart of Lemma 4.2 with $n_s = n$, $S = \mathcal{Z}_0(\Lambda^0)$, the fixed widths s, b, and $r = \sqrt{\mu}$. Let L_ω be a Lipschitz bound for $\omega = \nabla h$ near U.

Lemma 4.6 (Scaling)**.** *There are $M_1, \kappa, \mu_1, c > 0$, depending only on $(n, \nu, \mathcal{A}, \delta, \gamma, \tau)$, such that for every $\Lambda^0 \in \mathcal{L}_\delta$ satisfying (2.6) and every $0 < \mu \le \mu_1$ the following holds. Put*

$$r = \sqrt{\mu}, \qquad \vartheta = \frac{1}{2(\tau+1)}, \qquad N = \lfloor \kappa \mu^{-\vartheta} \rfloor, \qquad \epsilon = M_1 \sqrt{\mu}, \tag{4.12}$$

and $h_0 = h(\Lambda^0) + \omega(\Lambda^0) \cdot (\Lambda - \Lambda^0)$. Then $N \ge 1$, $\|X_{H_\mu - h_0}\|_3 \le \epsilon$, and (4.10) holds. In the transformed variables $(\tilde\Lambda, \tilde\lambda, \tilde z)$,

$$H_\mu \circ \Phi = h(\tilde\Lambda) + Y'(\tilde\Lambda, \tilde z) + R, \qquad \Big|\frac{d\tilde\Lambda}{dt}\Big|_\infty \le r\epsilon 2^{-N} \tag{4.13}$$

along every transformed solution in $\mathcal{D}_2$. With $T_\mu = \exp(c\mu^{-\vartheta})$, every portion of a solution of (2.1) lying in $\mathcal{D}_1$, of duration at most T_μ and starting at action Λ^0, satisfies $\|\Lambda(t) - \Lambda^0\|_\infty \le r/4$.

Proof. Shrink s, b so that f and its first derivatives are bounded by M on $\mathcal{D}_4$, and take μ_1 so small that $4r < r_*$ and $r \le 1$. On $\mathcal{D}_4$, the scaled field of $h - h_0$ has only angle components $(\omega(\Lambda) - \omega(\Lambda^0))/s$, of size at most $4L_\omega r/s$, while those of $X_{\mu f}$ are $\mu \partial_\lambda f/r$, $\mu \partial_\Lambda f/s$ and $\mu J \partial_z f/b$, of size at most $M\mu(1/r + 1/s + 1/b)$. The total $4L_\omega r/s + M\mu/r + O(\mu)$ is of order $\sqrt{\mu}$ at the balancing choice $r = \sqrt{\mu}$. Lemma 4.3 converts this into $\|X_{H_\mu - h_0}\|_3 \le M_1\sqrt{\mu}$, losing only the fixed strip between $\mathcal{D}_4$ and $\mathcal{D}_3$.

Condition (4.10) now reads $A_\gamma M_1 \sqrt{\mu}\, N^{\tau+1} \le q_{\mathsf{D}}$, so the admissible order is $N \asymp \mu^{-1/(2(\tau+1))}$. Precisely, choose κ with $A_\gamma M_1 \kappa^{\tau+1} \le q_{\mathsf{D}}/2$; since $\vartheta(\tau+1) = 1/2$, we get $A_\gamma \epsilon N^{\tau+1} \le A_\gamma M_1 \sqrt{\mu}\, \kappa^{\tau+1} \mu^{-1/2} \le q_{\mathsf{D}}/2$. Decrease μ_1 so that $\kappa\mu^{-\vartheta} \ge 2$, hence $N \ge \kappa\mu^{-\vartheta}/2$, and $\epsilon \le 1/8$. Lemma 4.5 gives Φ; with $Y' = h_0 - h + Y$ we obtain (4.13), since $d\tilde\Lambda/dt = -\partial_{\tilde\lambda} R$ has scaled size at most $\|X_R\|_2 \le \epsilon 2^{-N}$. Only the divisors $k \cdot \omega(\Lambda^0)$ were inverted; no arithmetic condition is imposed at later times.

Set $c = (\log 2)\kappa/4$, so $T_\mu = 2^{\kappa\mu^{-\vartheta}/4} \le 2^{N/2}$. As Φ is symplectic, Φ^{-1} maps a solution in $\mathcal{D}_1$ to a transformed solution in $\mathcal{D}_2$; adding the displacements of Φ^{-1} at time 0 and of Φ at time t to the drift (4.13), for $|t| \le T_\mu$,

$$\|\Lambda(t) - \Lambda^0\|_\infty \le \frac{r}{16} + r\epsilon|t|2^{-N} + \frac{r}{16} \le \frac{r}{8} + r\epsilon 2^{-N/2} \le \frac{r}{4}. \qquad \square$$

Remark 4.7 (Finite-cutoff normal form). The formal proof replaces Lemmas 4.3–4.6 by a finite-cutoff normal form of Pöschel type [10, p. 192] available in the formal library [1], fed with the data of Lemma 4.2: slow set the whole ball (4.4), fixed widths, cutoff $K \asymp \mu^{-\vartheta}$, nonresonance constant $\gamma K^{-\tau}/(2\pi)$, action radius $r = D_0\sqrt{\mu}$. Its smallness conditions hold since $\vartheta(\tau+1) = 1/2$, and it gives action drift at most $r/2$ up to a time $\ge \exp(c_1\mu^{-\vartheta})$ while the secular coordinates stay in the ball, which is the membership proved in Lemma 4.8. With $r/2$ in place of $r/4$, (4.14) still gives the margin $3\delta/4$ there.

4.4 First exit and the proof of Theorem B

The first-exit lemma is stated for an abstract chart, so that it also serves the spatial problem. A state is *prograde* if $q_i \neq 0$, $E_i < 0$ and all $H_i > 0$ (Lemma 3.3).

Lemma 4.8 (First exit)**.** *Assume the prograde states carry a real analytic symplectic chart (Λ, λ, z), $z \in \mathbb{R}^{2n_s}$, a diffeomorphism modulo the angle lattice, in which $H_\mu = h(\Lambda) + \mu f$ and $C = 1/2|z|^2$, and for which the conclusions of Lemmas 4.2 and 4.6 hold. Then there are $\mu_0, c, T_0, A_\Lambda, A_a, d_\star, d_{\mathrm{pair}} > 0$, depending only on $(n, \nu, \mathcal{A}, \delta, \gamma, \tau)$ and the chart, such that for $0 < \mu \leq \mu_0$ and every initial state satisfying (2.7), the maximal solution of (2.1) exists on $[-T_\mu, T_\mu]$, with T_μ as in (2.8); on this interval it is prograde, $\Lambda(t) \in \mathcal{Q}$, $C(t) \leq B(\Lambda(t)) - \delta/2$, $\|\Lambda(t) - \Lambda^0\|_\infty \leq A_\Lambda\sqrt{\mu}$, $\|a(t) - a(0)\|_\infty \leq A_a\sqrt{\mu}$, $|C(t) - C(0)| \leq nA_\Lambda\sqrt{\mu}$, and (3.6), $|q_i| \geq d_\star$ and $|q_i - q_j| \geq d_{\mathrm{pair}}$ $(i \neq j)$ hold.*

Proof. Constants. Take U, Ω_B, r_g as fixed above and μ_1, c from Lemma 4.6. Choose $\mu_0 \leq \mu_1$, before μ and the initial state, so that for $0 < \mu \leq \mu_0$, $r = \sqrt{\mu} \leq r_g$ and

$$nr + \Omega_B(r) \leq \delta/4, \qquad \Omega_B(r) \leq \delta/16. \tag{4.14}$$

Stopped interval. By Lemma 3.3, $C(0) \leq B(\Lambda^0) - \delta < B_\star(\Lambda^0)$ makes the initial state prograde, and $\Lambda^0 \in \mathcal{L}_\delta$. Let $\mathcal{J}$ be the open interval of times containing 0 on which the solution exists, is prograde, and satisfies the stopped conditions

$$\|\Lambda(t) - \Lambda^0\|_\infty < r, \qquad C(t) < B(\Lambda(t)) - \delta/2. \tag{4.15}$$

For $t \in \mathcal{J}$, the identity $C = 1/2|z|^2$ gives

$$\tfrac{1}{2}|z(t)|^2 = C(t) < B(\Lambda(t)) - \frac{\delta}{2} \leq B(\Lambda^0) + \Omega_B(r) - \frac{\delta}{2} \leq B(\Lambda^0) - \frac{7\delta}{16},$$

so $z(t) \in \mathcal{Z}_0(\Lambda^0)$ and the point lies in $\mathcal{D}_1$; this uses the whole ball (4.4) and only (4.15).

Improvement. Lemma 4.6 on $\mathcal{J} \cap [-T_\mu, T_\mu]$ gives $\|\Lambda(t) - \Lambda^0\|_\infty \leq r/4$, and then (3.8) with $r/4$ and (4.14) give $C(t) \leq B(\Lambda(t)) - 3\delta/4$.

Exit. Suppose $t_* = \sup \mathcal{J} \leq T_\mu$. If the maximal solution ends at t_*, this contradicts Lemma 3.6, as $nr + \Omega_B(r) \leq \delta/2$. Otherwise, since $E_i = -\nu_i^2/(2\Lambda_i^2) \leq -c_U < 0$ on $\mathcal{J}$, by continuity the state at t_* is elliptic, satisfies $\|\Lambda - \Lambda^0\|_\infty \leq r/4$ and $C \leq B(\Lambda) - 3\delta/4$, and is prograde as $H_i \geq 3\delta/4$ (Lemma 3.3); the open conditions then hold near t_*, a contradiction. Backward time is identical. Thus $[-T_\mu, T_\mu] \subset \mathcal{J}$; uniqueness holds because (2.1) is analytic. We may take $A_\Lambda = 1$, $T_0 = 1$, and $|C(t) - C(0)| \leq n\|\Lambda(t) - \Lambda^0\|_\infty$ by (3.3). Since $|a_i(\Lambda) - a_i(\Lambda^0)| = |\Lambda_i - \Lambda_i^0||\Lambda_i + \Lambda_i^0|/\nu_i^2$, $A_a = 2\Lambda_{\max}/\min_i \nu_i^2$ works. The separations come from Lemma 3.5 with $m = \delta/2$; by (3.7), $d_\star = a_{\min}\delta^2/(8\Lambda_{\max}^2)$. □

Proof of Theorem B. Here $\Lambda^0 \in \mathcal{L} \subset \mathcal{Q}$. By Lemma 3.3, $L \neq 0$; a reflection of the plane, applied to positions and momenta, is symplectic and leaves (2.1) and all constants unchanged, so we may assume $L > 0$, which is conserved. Prograde states then have all $L_i > 0$, the product chart of Lemma 4.1 covers them, and (4.3) gives $C = 1/2|z|^2$. Lemmas 4.2–4.6 and 4.8 give all conclusions except the osculating ones, with nothing assumed about the averaged Hamiltonian Y'. Lemma 3.1 reconstructs the Newtonian motion, and Lemma 4.9, on the compact set of Lemma 3.5 with $m = \delta/2$, gives the osculating conclusions once $\mu_0 \leq \mu_2$, $A_{\mathrm{osc}}\mu_0 \leq d_\star/2$ and $2A_{\mathrm{osc}}\mu_0 \leq d_{\mathrm{pair}}/2$. □

The velocity-osculating elements $E_i^{\mathrm{osc}}, a_i^{\mathrm{osc}}, \mathbf{e}_i^{\mathrm{osc}}, e_i^{\mathrm{osc}}$ of Section 2 use the heliocentric velocity $V_i = \dot{q}_i = p_i/\nu_i + \mu\sum_j p_j$ and the parameter $k_i = 1 + \mu\nu_i$.

Lemma 4.9 (Osculating elements)**.** *Let K be the compact set of Lemma 3.5 for a compact $U \subset \mathcal{Q}$ and $m > 0$, in dimension $d \in \{2,3\}$. There are $\mu_2, A_{\rm el}, A_{\rm osc} > 0$, depending only on K and ν, such that for $0 < \mu \le \mu_2$ and every state in K: $E_i^{\rm osc} < 0$, $a_i^{\rm osc} > 0$, $e_i^{\rm osc} < 1$, $a_i^{\rm osc}(1-e_i^{\rm osc}) \le |q_i| \le a_i^{\rm osc}(1+e_i^{\rm osc})$, and*

$$|a_i^{\rm osc} - a_i| + |\mathbf{e}_i^{\rm osc} - \mathbf{e}_i| \le A_{\rm el}\mu, \qquad |a_i^{\rm osc}(1 \pm e_i^{\rm osc}) - a_i(1 \pm e_i)| \le A_{\rm osc}\mu. \tag{4.16}$$

If also $A_{\rm osc}\mu \le d_\star/2$ and $2A_{\rm osc}\mu \le d_{\rm pair}/2$, then $a_i^{\rm osc}(1-e_i^{\rm osc}) \ge d_\star/2$ and $a_{i+1}^{\rm osc}(1-e_{i+1}^{\rm osc}) - a_i^{\rm osc}(1+e_i^{\rm osc}) \ge d_{\rm pair}/2$.

Sketch. On K, $|p_i| \le P$, $|q_i| \ge d_\star$, $E_i \le -1/(2a_{\max})$ and $e_i \le e_* < 1$. Then $|V_i - v_i| \le nP\mu$ and $|k_i - 1| \le \mu$, and expanding the definitions gives $|E_i^{\rm osc} - E_i| + |\mathbf{e}_i^{\rm osc} - \mathbf{e}_i| \le A_0\mu$. For small μ, $E_i^{\rm osc} \le -1/(4a_{\max})$ bounds $|a_i^{\rm osc} - a_i|$, and $|e_i^{\rm osc} - e_i| \le |\mathbf{e}_i^{\rm osc} - \mathbf{e}_i|$ (also at circular orbits) gives $e_i^{\rm osc} < 1$ and (4.16). These are the elements of the two-body orbit with parameter k_i through q_i with velocity V_i, so Lemma 3.2 with parameter k_i bounds $|q_i|$. The last claim follows from (3.6) and (4.16). □

5 The spatial problem

In space we need a chart regular at coplanar orbits whose slow variables measure the deficit against a fixed plane; that plane is aligned with L only along the actual solution. There are n fast angles and $2n$ slow pairs.

5.1 A regular spatial chart

Fix a unit vector $\mathbf{N} \in \mathbb{R}^3$ and coordinates in which $\mathbf{N} = (0,0,1)$. For one planet with parameter $\nu > 0$ the chart has variables $(\Lambda, \lambda, x, y, u, v)$, with $\lambda \in \mathbb{T}$ and

$$\Lambda > \Gamma + Z, \qquad \Gamma = \tfrac{1}{2}(x^2+y^2), \qquad Z = \tfrac{1}{2}(u^2+v^2). \tag{5.1}$$

Let (Q,P) be the planar chart of Lemma 4.1 at (Λ,λ,x,y), embedded in the first two coordinate directions, and put

$$t = \frac{(u,-v,0)}{2\sqrt{\Lambda-\Gamma}}, \qquad c_t = \sqrt{1-|t|^2}, \qquad R = (1-2|t|^2)\,\mathrm{Id} + 2\,tt^{\mathsf{T}} + 2c_t\,[t]_\times, \qquad (q,p) = (RQ, RP), \tag{5.2}$$

where $[t]_\times w = t \times w$; as $|t|^2 = Z/(2(\Lambda-\Gamma)) < 1/2$, the roots are positive and analytic.

Lemma 5.1 (Regular spatial Poincaré chart)**.** *For every $\nu > 0$, (5.1)–(5.2) define a real analytic diffeomorphism, with λ read modulo 2π, onto the states $(q,p) \in \mathbb{R}^3 \times \mathbb{R}^3$ with negative energy and $(q \times p)\cdot \mathbf{N} > 0$. It carries $dp \wedge dq$ to $d\Lambda \wedge d\lambda + dx \wedge dy + du \wedge dv$, it is regular at circular and at coplanar orbits, and*

$$\frac{|p|^2}{2\nu} - \frac{\nu}{|q|} = -\frac{\nu^3}{2\Lambda^2}, \qquad |q \times p| = \Lambda - \Gamma, \qquad (q \times p)\cdot \mathbf{N} = \Lambda - \Gamma - Z. \tag{5.3}$$

For n planets, the fixed-plane deficit of the product chart is

$$C_{\mathbf{N}} = \sum_i \Lambda_i - L\cdot\mathbf{N} = \tfrac{1}{2}\sum_i \big(x_i^2 + y_i^2 + u_i^2 + v_i^2\big). \tag{5.4}$$

Sketch. One checks $R \in SO(3)$ from $[t]_\times^2 = tt^{\mathsf{T}} - |t|^2\mathrm{Id}$, and $R\mathbf{N}$ has third component $1 - Z/(\Lambda - \Gamma)$. As $Q \times P = (\Lambda - \Gamma)\mathbf{N}$ (Lemma 4.1), $q \times p = (\Lambda - \Gamma)R\mathbf{N}$; this gives (5.3), and summing over the planets gives (5.4). Where the eccentricity and Z are positive, write $(x, y) = \sqrt{2\Gamma}(\cos\varpi, -\sin\varpi)$ and $(u, v) = \sqrt{2Z}(\cos\Omega, -\sin\Omega)$; then R is a rotation about the line of nodes conjugated by the rotation by Ω about $\mathbf{N}$. A varying rotation adds $(q\times p)\cdot\boldsymbol{\theta}$ to $p\cdot dq$, where $\boldsymbol{\theta}$ is the infinitesimal rotation vector, and the line of nodes is orthogonal to $q \times p$, so $p \cdot dq = \Lambda\, d\lambda + \Gamma\, d(-\varpi) + Z\, d(-\Omega) + d(\cdot)$, with $\Gamma\, d(-\varpi) = \frac{1}{2}(x\, dy - y\, dx)$ and $Z\, d(-\Omega) = \frac{1}{2}(u\, dv - v\, du)$. This gives the form on a dense set, and analyticity extends it. The inverse is explicit: $q \times p$ determines $\Lambda - \Gamma$, u, v, hence R, and Lemma 4.1 inverts the planar part [2]. □

5.2 The spatial domain

Lemma 5.2 (Spatial domain)**.** *Lemma 4.2 holds in space, for the product of the charts of Lemma 5.1, with $2n$ slow pairs $z = (x_i, y_i, u_i, v_i)_i \in \mathbb{R}^{4n}$ and the whole ball*

$$\mathcal{Z}_0(\Lambda^0) = \left\{ z \in \mathbb{R}^{4n} : \tfrac{1}{2}|z|^2 \le B(\Lambda^0) - \delta/4 \right\}; \tag{5.5}$$

its constants do not depend on $\mathbf{N}$. Every state allowed by (2.7) lies in this domain once $\mathbf{N}$ is aligned with its total angular momentum.

Proof. For a real point of (5.5) and a real Λ with $\|\Lambda - \Lambda^0\|_\infty \le 4r_g$, using $\Omega_B(4r_g) \le \delta/16$ and $B \le B_\star$,

$$\Gamma_i + Z_i \le \tfrac{1}{2}|z|^2 \le B(\Lambda^0) - \delta/4 \le B(\Lambda) - 3\delta/16 \le \Lambda_i - 3\delta/16.$$

Hence $\Lambda_i - \Gamma_i$, $L_i \cdot \mathbf{N} \ge 3\delta/16$ and $|t_i|^2 < 1/2$ with fixed margins. Moreover $\Lambda_i D(e_i) = \Lambda_i - |L_i| \le \Lambda_i - L_i \cdot \mathbf{N}$, so by (5.4) $\sum_i \Lambda_i D(e_i) \le C_{\mathbf{N}} = \frac{1}{2}|z|^2 \le B(\Lambda) - 3\delta/16$. Thus Lemma 3.5 applies, with $m = 3\delta/16$, at every point of the ball, also where L is not parallel to $\mathbf{N}$: alignment is not a restriction on the analytic domain. The rest of the proof of Lemma 4.2 applies verbatim in $6n$ coordinates (compactness, holomorphic extension of (5.2) and Lemma 4.1 with these margins, gluing, and the bilinear denominators $(q_i - q_j) \cdot (q_i - q_j)$). Scalar products and potentials are invariant under a simultaneous rotation of all positions and momenta, so the constants do not depend on $\mathbf{N}$.

For a state allowed by (2.7), $L \neq 0$ by Lemma 3.3. Fix $\mathcal{R} \in SO(3)$ with $\mathcal{R}L = |L|\mathbf{N}$; the map $(q, p) \mapsto (\mathcal{R}q, \mathcal{R}p)$ is symplectic and preserves H_μ and all elements, and one rotation is used per solution. Along the solution L is conserved, so $C_{\mathbf{N}} = C$, and $L_i \cdot \mathbf{N} > 0$ by Lemma 3.3 while $C < B_\star(\Lambda)$. □

5.3 Proofs of Theorem C and Corollary D

Proof of Theorem C. After the rotation of Lemma 5.2, the product chart is symplectic, $H_\mu = h + \mu f$, and $\frac{1}{2}|z|^2 = C$ along the solution. Lemmas 4.3–4.6 hold verbatim with $n_s = 2n$, which changes only constants, not exponents; all small divisors are $k \cdot \omega(\Lambda^0)$, $k \in \mathbb{Z}^n$. Lemmas 4.8, 3.5 (dimension-free), 4.9 and 3.1 give all conclusions, and undoing the rotation changes no inequality. □

Proof of Corollary D. $\tau = n - 1/2 > n - 1$ is admissible in Theorems B and C, and $2(\tau + 1) = 2n + 1$. □

6 Resonant mean motions

The Diophantine condition enters the proofs of Theorems B and C only through the stopped drift estimate of Lemma 4.6. We replace it by Nekhoroshev's theorem for a convex Hamiltonian with slow variables. The exponents and the mechanism are Nekhoroshev's: the Kepler Hamiltonian is convex [7, §1.18], and he proved the absence of collisions over exponentially long times, with the same exponents, on his domains of planetary motions [7, Theorem 12.3, pp. 55–56]. What is added is the sharp hypothesis $C(0) \le B(\Lambda(0)) - \delta$ on the whole curved sublevel.

6.1 The input theorem

Write points of $\mathbb{R}^{2(n+n_s)}$ as $(\theta, \xi; I, \eta)$, with fast angles θ, fast actions $I \in \mathbb{R}^n$ and slow pairs $(\xi, \eta) \in \mathbb{R}^{n_s} \times \mathbb{R}^{n_s}$, and the form $\sum dI_i \wedge d\theta_i + \sum d\eta_k \wedge d\xi_k$. For a set X and $d > 0$ let $X \ominus d = \{x \in X : \{\|y - x\|_\infty \le d\} \subset X\}$; $\mathfrak{a}_n, \mathfrak{b}_n$ are as in Theorem E.

Proposition 6.1 (Nekhoroshev, definite case with slow variables)**.** *Let $n \ge 2$, $n_s \ge 1$, let $G \subset G' \subset \mathbb{R}^n$ be open and convex, $S \subset \mathbb{R}^{2n_s}$ open and $\rho_* > 0$, with the closed ρ_*-ball about every point of G contained in G'. Let h be holomorphic on $G'_{\rho_*} = \{I \in \mathbb{C}^n : \mathrm{dist}_\infty(I, G') < \rho_*\}$, real on real points, with $\|D^2 h\| \le M_2$ there, and let $g, m_h > 0$ and $\sigma \in \{1, -1\}$ be such that $\|\nabla h(I)\|_\infty \ge g$ and $\sigma D^2 h(I)(v, v) \ge m_h |v|^2$ for all real $I \in G'_{\rho_*}$ and $v \in \mathbb{R}^n$. Then there exists $\mu_0 > 0$, depending only on $(n, n_s, G, G', S, \rho_*, h, g, m_h, M_2)$, with the following property.*

Let $0 < \varepsilon < \mu_0$, and let P be holomorphic on the set F of $(\theta, \xi; I, \eta) \in \mathbb{C}^{2(n+n_s)}$ with $\mathrm{Re}\, I \in G'$, $(\mathrm{Re}\,\xi, \mathrm{Re}\,\eta) \in S$ and all imaginary parts of modulus $< \rho_*$, real on real points, 2π-periodic in each θ_i, with $|P| \le 1$ on F. Let $C > 0$ and let (θ, ξ, I, η) solve Hamilton's equations for $h(I) + \varepsilon P$ on $[0, C]$, with $I(t) \in G$ there and $I(0) \in G \ominus \varepsilon^{\mathfrak{b}_n}$. Then $\|I(t) - I(0)\|_\infty \le \frac{1}{2}\varepsilon^{\mathfrak{b}_n}$ for every $t \in [0, \min\{C, \varepsilon^{-1} \exp(\varepsilon^{-\mathfrak{a}_n})\}]$ such that $(\xi(u), \eta(u)) \in S \ominus \varepsilon^{\mathfrak{b}_n}$ for all $u \in [0, t]$.

The threshold precedes P, C and the solution. This is [7, Theorem 4.4, p. 30], with $\sup|H_1|$ as allowed by [7, Remark 4.5.A], specialized to a definite Hessian, which is quasiconvex with all steepness indices one [7, §1.11, p. 9]; then [7, (4.1), (4.4), (4.5)] give $\zeta = n(n-1)/2$ and the exponents $\mathfrak{a}_n, \mathfrak{b}_n$. It is stated in a slightly weaker form: analyticity on an enlargement G' of G, erosions in the maximum norm (stronger than the Euclidean ones of [7, §4.2]), the conclusion $\le \frac{1}{2}\varepsilon^{\mathfrak{b}_n}$ in the maximum norm, and μ_0 depending on the domains. It is proved in this form in Lean (Appendix A).

6.2 A finite cover

The threshold of Proposition 6.1 is not quantitative in $G' \times S$, while the slow radius of (2.7) varies with $B(\Lambda)$. We use finitely many cells, each with a slow ball at its own sharp level. Let $n_s = n$ in the plane and $n_s = 2n$ in space, and take U, r_g, Ω_B as in Section 4 and r_*, s, b, M from Lemma 4.2 or 5.2, renaming the slightly smaller domain there so that f is holomorphic, real, 2π-periodic in λ and bounded by M on the domain (4.5) centred at any $\Lambda^0 \in \mathcal{L}_\delta$. Put $h(\Lambda) = -\sum_i \nu_i^3/(2\Lambda_i^2)$ and

$$w = \min\{r_*, 4s, 4b\}, \qquad \eta = w/4, \qquad \rho_* = w/8, \qquad B_0 = \max\{M, 1\}, \qquad P = f/B_0.$$

Lemma 6.2 (Finite cover)**.** *There are $c_1, \dots, c_J \in \mathcal{L}_\delta$ with $\mathcal{L}_\delta \subset \bigcup_j \{\|\Lambda - c_j\|_\infty < \eta/2\}$ such that the cells $G_j = \{\|\Lambda - c_j\|_\infty < \eta\}$, $G'_j = \{\|\Lambda - c_j\|_\infty < 2\eta\}$ and $S_j = \{z \in \mathbb{R}^{2n_s} : \frac{1}{2}|z|^2 < B(c_j) - 5\delta/16\}$ satisfy, with constants independent of j:*

(i) $(G_j, G_j', S_j, \rho_*, \pm h)$ *satisfy the hypotheses of Proposition 6.1 with* $\sigma = \mp 1$*, and* $\pm P$ *satisfy them on the domain* F_j *built from* (G_j', S_j, ρ_*)*, on whose real points the chart Hamiltonian is* $h + B_0\mu P$*;*

(ii) if $\Lambda^0 \in \mathcal{L}_\delta$*,* $\|\Lambda^0 - c_j\|_\infty < \eta/2$*,* $\Lambda \in U$ *and* $\|\Lambda - \Lambda^0\|_\infty \le 4r_g$*, then* $B(\Lambda) \le B(c_j) + \delta/8$*;*

(iii) there is $\varepsilon_D > 0$ *with* $\{\frac{1}{2}|z|^2 < B(c_j) - 3\delta/8\} \subset S_j \ominus r$ *for all* j *and* $0 < r < \varepsilon_D$*.*

Proof. The cover exists by compactness, and $\eta + \rho_* < 2\eta$. (i) Points of $(G_j')_{\rho_*}$ and of F_j have actions within $2\eta + \rho_* < w \le r_*$ of c_j, so with real parts in U; there h is holomorphic and real, $D^2h = \mathrm{diag}(-3\nu_i^3/\Lambda_i^4)$ is bounded, and at real points $\omega_i \ge g$ and $-D^2h(v,v) \ge m_h|v|^2$ for constants $g, m_h > 0$ depending on U and w. Points of F_j also have angles with imaginary parts below $\rho_* < 4s$ and slow coordinates within $\rho_* < 4b$ of $S_j \subset \mathcal{Z}_0(c_j)$, so F_j lies in the domain (4.5) centred at c_j, and the splitting $H_\mu = h + \mu f$ gives the last claim. (ii) As $\eta/2 \le r_g$, $B(\Lambda) \le B(\Lambda^0) + \Omega_B(4r_g) \le B(c_j) + 2\Omega_B(4r_g) \le B(c_j) + \delta/8$. (iii) With $A_j = B(c_j) - 5\delta/16$ and $A_j' = B(c_j) - 3\delta/8 \ge 5\delta/8$: if $\frac{1}{2}|z|^2 < A_j'$ and $\|z' - z\|_\infty \le r$, then $|z'| < \sqrt{2A_j'} + \sqrt{2n_s}\,r$; take $\varepsilon_D = \min_j(\sqrt{2A_j} - \sqrt{2A_j'})/\sqrt{2n_s}$. □

6.3 Proof of Theorem E

Constants. Proposition 6.1 gives, for each j, thresholds $\mu_j^\pm$ for $\pm h$. Let $r_1 > 0$ be such that $nr + \Omega_B(r) \le \delta/4$ for $r \le r_1$. Choose $\varepsilon_0 \in (0,1]$ with $\varepsilon_0 \le \min_j \mu_j^\pm$, $\varepsilon_0^{\mathfrak{b}_n} \le \min\{\eta/4, r_1, r_g\}$ and $\varepsilon_0^{\mathfrak{b}_n} < \varepsilon_D$, and let $\mu_0 \le \varepsilon_0/(2B_0)$ satisfy also the conditions of Lemma 4.9. For $0 < \mu \le \mu_0$ put $\varepsilon = B_0\mu < \varepsilon_0$, $r = \varepsilon^{\mathfrak{b}_n}$ and $T = \exp(\varepsilon^{-\mathfrak{a}_n}) = T_0 \exp(c\mu^{-\mathfrak{a}_n})$, with $T_0 = 1$, $c = B_0^{-\mathfrak{a}_n}$; all this precedes the initial state.

Stopped drift. Lemma 4.8 uses the normal form only through this estimate: if $|t| \le T$ and for all u between 0 and t the state is prograde and satisfies (4.15), then $\|\Lambda(t) - \Lambda^0\|_\infty \le r/2$. To prove it, orient or rotate as in Theorems B and C and pass to the chart, where $\frac{1}{2}|z|^2 = C$. As $\Lambda^0 \in \mathcal{L}_\delta$, choose j with $\|\Lambda^0 - c_j\|_\infty < \eta/2$. Since $r \le \eta/4$, $\Lambda^0 \in G_j \ominus r$ and $\Lambda(u) \in G_j$. By Lemma 6.2(ii),

$$\tfrac{1}{2}|z(u)|^2 = C(u) < B(\Lambda(u)) - \delta/2 \le B(c_j) - 3\delta/8,$$

so $z(u) \in S_j \ominus r$ by Lemma 6.2(iii); the bound $C < B(\Lambda^0) - \delta/4$ of Lemma 4.8 would leave no room for this erosion. The orbit thus stays in the real points of F_j, where it solves Hamilton's equations for $h + \varepsilon P$. For $t > 0$, as $\varepsilon \le 1$, $t \le \exp(\varepsilon^{-\mathfrak{a}_n}) \le \varepsilon^{-1}\exp(\varepsilon^{-\mathfrak{a}_n})$, and Proposition 6.1 with $C = t$ gives $\|\Lambda(t) - \Lambda^0\|_\infty \le \frac{1}{2}\varepsilon^{\mathfrak{b}_n} = r/2$. For $t < 0$, the time-reversed orbit on $[0, -t]$ solves Hamilton's equations for $-h - \varepsilon P$; use μ_j^-.

Conclusion. With r and T as above, $nr + \Omega_B(r) \le \delta/4$ and $r \le r_g$, and the improvement to $r/2$ still gives the margin $3\delta/4$, so the proof of Lemma 4.8 applies unchanged. It gives existence and uniqueness on $[-T, T]$, the margin $\delta/2$, the separations, and the drift bounds with $A_\Lambda = B_0^{\mathfrak{b}_n}/2$; Lemmas 4.9 and 3.1 give the rest. In space, $n_s = 2n$ and h depends only on the n fast actions, so the exponents are unchanged. Table 1 compares the results. □

7 Limitations and open questions

The results are asymptotic. The proofs of Theorems B–E do not compute μ_0 or the other constants. They come from compactness: the widths of the complex domain around the whole sublevel, the bound on the complexified perturbation there, the modulus of continuity of B, and,

	Theorems B, C	Theorem E
arithmetic hypothesis	Diophantine (γ, τ) at $\omega(\Lambda(0))$	none
constants depend on	$(n, \nu, \mathcal{A}, \delta, \gamma, \tau)$	$(n, \nu, \mathcal{A}, \delta)$
time	$T_0 \exp(c\mu^{-1/(2(\tau+1))})$	$T_0 \exp(c\mu^{-\mathfrak{a}_n})$
action, axis, AMD drift	$O(\sqrt{\mu})$	$O(\mu^{\mathfrak{b}_n})$
other conclusions	sharp B, margin $\delta/2$	identical

Table 1: The Diophantine and the resonance-tolerant results; neither implies the other.

for Theorem E, the thresholds of Proposition 6.1, which are not explicit either. The theorems say nothing about a given value of μ, and no statement about a real planetary system follows from them.

Real masses. An explicit version would need rigorous lower bounds for the complex widths near the boundary of the sublevel, where perihelia and mutual distances approach their minima, rigorous bounds for the complexified perturbation on the whole enlarged slow set, and, for Theorems B and C, control of the small divisors up to a cutoff. With generic majorants one should expect admissible mass ratios many orders of magnitude below planetary values. Tailored estimates do better, but remain far from planetary masses: for a planar Sun–Jupiter–Saturn model near the 5:2 resonance, [3, §5.1] reaches times comparable to the age of the Solar System for perturbations of size about 10^{-11} in their normalization, while the Jupiter–Sun mass ratio is about 10^{-3}. Nor can observations establish a Diophantine condition to all orders, since every interval of mean motions contains resonances of high order; Theorem E removes this hypothesis, but an explicit version would still have to control resonances up to a finite order.

The Solar System. We make no claim about the Solar System. Laskar and Petit find it AMD-unstable, because of the AMD carried by the giant planets [6, §5.4.1], so the hypothesis of our theorems fails for the full eight-planet Newtonian model, whatever the constants. The model (2.1) also omits all non-Newtonian effects.

Open questions.

1. *Explicit constants.* Can the proof be made quantitative, perhaps computer-assisted with a finite Fourier cutoff and rigorous divisor bounds, for a published model of a system with a large AMD margin?

2. *Exponents.* The exponents of Theorem E are those of Nekhoroshev's general theorem, as in [7, Theorem 12.3]. Can they be improved on the sharp sublevel, using the convexity of the Kepler Hamiltonian or the structure of specific resonances?

3. *Other thresholds.* What are the analogues of our theorems for the weaker notion of AMD-stability of [6, §5.3], which allows a collision of the innermost planet with the star?

4. *Longer times.* Does a fixed margin below the strong threshold give stability beyond exponentially long times?

5. *Earlier work.* Niederman [8] proved exponential stability results in the planetary problem. We have not inspected that paper; a precise comparison of hypotheses, domains and exponents remains to be made.

A Formal verification

Theorems A–C and Corollary D are formally verified in Lean 4 [5] (version 4.35.0) with the Mathlib library [11], in the repository [1] at commit `a994efdf`. The formal proof of Theorem E is in preparation in the same repository. This appendix records what the formal proofs establish, what they do not, and where the statements are.

A.1 What is verified and what is not

The Lean statements are written in the model of Section 2: the exact Hamiltonian (2.1) in heliocentric positions and scaled canonical momenta, the Kepler energies, axes, actions and eccentricity vectors computed from (q, p), the AMD $C = \sum_i \Lambda_i - |L|$, and the thresholds (2.4) defined as minima over the action chamber. The conclusions include the existence and uniqueness of the solution of Hamilton's equations on $[-T, T]$, and the fact that its barycentric reconstruction solves Newton's equations with masses $(1, \mu\nu_1, \dots, \mu\nu_n)$. All constants are quantified before μ and the initial state, exactly as in the paper. The general theorems do not assume $\max_i \nu_i = 1$; the normalized forms are corollaries.

Lean checks that each stated conclusion follows from the stated hypotheses and the definitions, using Mathlib and the axioms listed below. It does not check:

- that the Lean definitions express the intended physical and mathematical notions (heliocentric canonical coordinates, the Kepler elements, the AMD, the thresholds, the osculating elements). This was argued by comparing every definition and statement with the mathematical text, in reviews by readers other than the writers of the formal proofs;
- the identification of B_{ij} with the critical AMD of Laskar and Petit (Lemma 3.4), which is not formalized;
- the proofs of Lemmas 4.3–4.6: the formal proofs of Theorems B and C use instead the finite-cutoff normal form already available in the library (Remark 4.7);
- the literature citations and the prose of this paper.

Proposition 6.1, the input of Theorem E, is itself formally proved in the same library; its statement was fixed in advance, with [7, Theorem 4.4] as its source. The file containing that proof, at commit `a994efdf`, contains no `sorry`; the formal proof of Theorem E depends on it.

A.2 Statements

Table 2 lists the formal statements of the main results. Names in the planar files are in the namespace `KAM.Planetary.StrongAMD`, names in the spatial files in `KAM.Planetary.Spatial`, and the name of Proposition 6.1 in `KAM.Nekhoroshev`. Paths are relative to `KAM/Planetary/` except for the last row.

Several lemmas have formal counterparts that can be read independently; Table 3 lists the principal ones. They formalize the content of the lemma, sometimes split differently.

A.3 Axioms and status

As reported by the project's build checks at commit `a994efdf`, the declarations of Table 2 for Theorems A–C and Corollary D depend only on the axioms `propext`, `Classical.choice` and

Result	Lean declaration (file and line)
Thm. A	`lp0_planar_barrier` (`StrongAMD/PlanarSafety/Main.lean:255`)
spatial	`spatialStrongBarrier_physical`, `exists_spatialBarrierCompact` (`Spatial/Geometry.lean:83, 252`)
Thm. B	`lp1_planar_strongAMD_safety`, `lp1_planar_strongAMD_safety_normalized` (`StrongAMD/PlanarSafety/Main.lean:44, 137`)
Thm. C	`lp2_spatial_strongAMD_safety`, `lp2_spatial_strongAMD_safety_normalized` (`Spatial/SpatialSafety/Main.lean:43, 153`)
Cor. D	`lp1_planar_strongAMD_safety_halfInteger` (`StrongAMD/PlanarSafety/Main.lean:190`), `lp2_spatial_strongAMD_safety_halfInteger` (`Spatial/SpatialSafety/Main.lean:210`)
Thm. E	`lp3_planar_strongAMD_safety`, `lp3_spatial_strongAMD_safety`, and their normalized forms (in preparation)
Prop. 6.1	`partialConvex_nekhoroshev` (`KAM/Frozen/Nekhoroshev/Theorems/PartialConvexNekhoroshev.lean:48`)

Table 2: Formal statements of the main results, at commit `a994efdf`.

`Quot.sound`, the standard axioms of Lean and Mathlib; in particular they use no `sorry`. The same is reported for the development version of the formal proof of Theorem E.

Theorems A–C and Corollary D are proved, and their formal statements and proofs have been independently reviewed. Theorem E is proved on a development version; its independent review is in progress. The exact statement texts of all results are under final review.

A.4 Use of AI and formal tools

The proofs in this paper, their Lean formalization and a first draft of the text were developed with substantial use of large language model systems, working under the direction of the authors. Every formal proof was checked by the Lean kernel. Statements, proofs and the correspondence between the paper and the formal statements were reviewed by independent model-based audits and by the authors. The authors take full responsibility for the content of the paper.

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[3] Santiago Barbieri and Laurent Niederman, *Sharp Nekhoroshev estimates for the three-body problem around periodic orbits*, J. Differential Equations **268** (2020), 3749–3780, Locators refer to arXiv:1810.05987v1.

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Lemma	Lean declarations (file and line)
3.4	`exists_pairThreshold_attained` (`StrongAMD/Threshold/Pair.lean:97`), `continuousOn_strongThreshold` (`StrongAMD/Threshold/Strong.lean:120`)
3.6	`strongAMD_le_of_tube`, `maximal_global_of_barrier` (`StrongAMD/Continuation.lean:43, 72`)
4.1	`KAM.Planetary.poincare_symplectic` (`Poincare/Symplectic.lean:642`), `KAM.Planetary.poincare_image` (`Poincare/Bijective.lean:320`), `lpChart_strongAMD` (`StrongAMD/PlanarDictionary.lean:156`)
4.2	`lemma8_complete` (`StrongAMD/AnalyticDomain/Bounds.lean:185`)
4.8	`strongFirstExit_of_stopped_drift` (`StrongAMD/Confinement.lean:170`)
4.9	`osculating_barrier` (`StrongAMD/Osculating.lean:268`)
5.1	`spatialPoincare_image` (`Spatial/Chart/Image.lean:180`), `spatialPoincare_symplectic` (`Spatial/Chart/Symplectic.lean:256`), `spatialFixedPlaneDeficit_spatialPoincareProd` (`Spatial/Product/Deficit.lean:110`)
5.2	`lemma18_complete` (`Spatial/AnalyticDomain/Bounds.lean:191`)

Table 3: Formal counterparts of lemmas, at commit `a994efdf`.

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